

TCS349: Responsible and Explainable AI

Unit 2, Lecture 7

Privacy in AI: Collection, Consent, Profiling & Misuse

How personal data enters AI systems, what meaningful consent looks like, and where profiling and misuse turn ordinary data collection into a privacy risk.

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Data Collection



Consent



Profiling



Data Misuse

1. Why Privacy Matters in AI Systems

AI systems are often trained and operated on large volumes of personal data, making privacy a central, not peripheral, concern.

Privacy failures can cause direct harm to individuals - financial loss, discrimination, reputational damage or unwanted exposure.

Privacy is closely linked to the transparency and accountability principles from earlier in this unit - each depends on knowing what data is used and why.

Regulators increasingly treat privacy as a legal requirement, not just good practice, for any AI system using personal data.

2. Types of Personal Data

Type	Example
Direct Identifiers	Name, email address, phone number, government ID
Indirect Identifiers	ZIP code, birth date, job title, device ID
Sensitive Categories	Health data, religion, sexual orientation, biometric data

3. Data Collection - Sources & Risks

Personal data enters AI systems through direct input (forms, sign-ups), passive collection (sensors, cookies, location tracking) and third-party sources (data brokers, public records).

Passive and third-party collection often happens without the individual's active awareness.

Collecting more data than necessary increases risk without necessarily improving system performance.

The scale and permanence of digital data collection make privacy harms harder to reverse than in the past.

4. Consent - What Meaningful Consent Requires

Meaningful consent must be informed, specific and freely given - not buried in unread terms and conditions.

Informed: the person understands what data is collected and why, in plain language.

Specific: consent is given for a particular use, not an open-ended, unlimited one.

Freely given: the person can decline without losing access to an essential service.

5. Consent in Practice - Good vs Poor Examples

Poor Consent Practice	Good Consent Practice
One bundled "accept all" checkbox for every use of data	Separate toggles for each distinct use of data
Legal jargon that an average user cannot understand	Plain-language explanation of what is collected and why
Service is unusable unless all data sharing is accepted	Core service still works if optional sharing is declined

6. Profiling - What It Is & Why It's Risky

Profiling is the automated analysis of personal data to infer traits, preferences or predictions about an individual.

It can infer sensitive information - such as health status or political views - even from data that seems unrelated or anonymous.

Profiling can lead to differential treatment: different prices, offers or opportunities based on an inferred profile rather than a person's actual choices.

Because profiling operates on inferred data, the person profiled is often unaware of how they are being categorised.

7. Purpose Limitation - Using Data Only as Intended

Purpose limitation means data collected for one stated purpose should not be reused for a different, incompatible purpose without new consent.

It directly opposes a common temptation: reusing an existing dataset for a new AI application simply because the data is already available.

Violating purpose limitation breaks the trust that consent was originally built on, even if no law is technically broken.

This principle is central to why datasheets, introduced in Lecture 12, document a dataset's intended and recommended uses.

8. Data Misuse - Common Failure Patterns

Scope creep: data collected for one purpose gradually gets used for unrelated analytics or profiling.

Excessive retention: data is kept far longer than needed, increasing exposure if a breach occurs.

Unauthorised sharing: data is passed to third parties without the individual's knowledge or consent.

Re-identification: supposedly anonymised data is combined with other sources to identify individuals again.

9. Case Example: Privacy Failure in Practice

A fitness-tracking app shared aggregated user location data publicly to show popular exercise routes.

The aggregated data inadvertently revealed the locations of sensitive sites, including military bases, through unusual activity patterns.

No individual was named, yet the pattern of behaviour itself exposed information users never intended to share.

The case shows that privacy risk does not require identifying a person by name - patterns and inference can be enough.

10. Privacy by Design - Building It In Early

Privacy by design means privacy safeguards are built into a system from the start, not added after a problem is discovered.

Core practices include data minimisation, clear consent flows, and limiting who inside an organisation can access personal data.

Building privacy in early is consistently cheaper and more effective than retrofitting it after deployment.

Lecture 17 continues this thread with the specific techniques used to implement privacy by design: minimisation, anonymisation and access control.